Hydrostatic pressure the lock gates must resist

1. Data and assumptions

- Opening 4.2 m wide, gate height 6.0 m, head of water 2.0 m to 5.5 m (brief). Design for the worst case: 5.5 m on the chamber side and the downstream side empty.
- Fresh canal water: density $\rho = 1000$ kg/m³, $g = 9.81$ m/s², so $\rho g = 9.81$ kN/m³. The water is still (hydrostatic): no waves or surge.
- Mitre gates: two leaves 2460 mm long (pivot to mitre) meeting at 1 in 3 ($\theta = 18.4°$). Each leaf takes the water on its own length.
- Steel S355 (yield 355 MPa, $E = 210$ GPa). Pintle 316 stainless (yield 250 MPa used). Self weight from the CAD model.
- Gates only move when the levels are within 50 mm of each other (control interlock).

2. Pressure at depth

$p = \rho g h$. At the cill with 5.5 m of water: $p = 1000 \times 9.81 \times 5.5 =$ **53.95 kPa**. With the minimum 2.0 m head: $p = 19.62$ kPa. The pressure is zero at the surface and grows in a straight line to the cill, so the pressure diagram is a triangle.

3. Force on the gates

Quantity	Working	Result
Force per metre width (5.5 m)	$F = \tfrac{1}{2}\rho g h^2 = 0.5 \times 9.81 \times 5.5^2$	**148.4 kN/m**
Force per metre width (2.0 m)	$0.5 \times 9.81 \times 2.0^2$	19.6 kN/m
Height of the centre of pressure	$h / 3 = 5.5 / 3$, above the cill	1.83 m
Force on one leaf (square to the leaf)	$P = 148.4 \times 2.46$	**365 kN**
Total along the lock (both leaves)	148.4×4.2	623 kN
Thrust at the mitre and at each quoin	$R = P / (2 \sin\theta) = 365 / (2 \times 0.3162)$	**577 kN**

The two leaves act as a three pin arch: the water pushes each leaf against the other at the mitre and back into the quoin. That is why a mitre gate points upstream: the higher the water, the tighter the mitre and quoin seals.

4. Skin plate panel (the plate that holds back the water)

Bottom panel, 580 × 630 × 12 mm, fixed on all four edges by the girders, the heel post and a stiffener. Pressure at its centre (475 mm above the cill): $p = 9.81 \times (5.5 - 0.475) = 49.3$ kPa $= 0.0493$ MPa.

Check (Roark, flat plate, all edges fixed, a/b = 1.09)	Hand	FEA
Bending stress at the middle of the long edge: $\sigma = \beta p b^2 / t^2$, $\beta = 0.340$	39.2 MPa	—
Von Mises at that point ($\sigma_2 = 0.3\ \sigma_1$)	34.8 MPa	34.1 MPa
Deflection at the centre: $y = \alpha p b^4 / (E t^3)$, $\alpha = 0.0160$	0.25 mm	0.25 mm
Safety factor on yield (355 MPa)	10.2	10.4

The hand calculation and the FEA agree to within 2 % on stress and 1 % on deflection. **Panel OK.**

5. Bottom girder (worst girder)

Carries the water on the strip from the cill to half way to the next girder (0 to 0.475 m): $w = \rho g (h\,z - z^2/2)$ = **24.5 kN/m**. Span between the heel and mitre posts L = 1.76 m. Treated as simply supported, which ignores the arch action and so is on the safe side: $M = wL^2/8$ = 9.49 kNm.

Section: 400 mm of skin plate acting with the girder, web 20 mm, flange 200 × 20 mm. A = 17160 mm², neutral axis 13 mm from the leaf centre, $I = 534 \times 10^6$ mm⁴. $\sigma = M\,y / I = 9.49 \times 10^6 \times 238 / 534 \times 10^6$ = **4.2 MPa**, safety factor 84. **OK.** The girders are governed by stiffness, the axial thrust from the mitre (577 kN shared by eight girders) and boat impact, not by bending stress.

6. Leaf weight, hinges and lifting

Quantity	Working	Result
Leaf mass (CAD model: every part that turns with the leaf)	steel 7850 kg/m³, oak 750 kg/m³	**8.89 t**
Weight on the bottom pintle	W = mg	87.2 kN
Centre of gravity from the pivot	from the CAD model	1290 mm
Sideways load on each hinge	H = W x / h = 87.2 × 1.290 / 5.78	19.5 kN
Pintle direct stress (150 mm pin)	W / (πd²/4)	4.9 MPa
Pintle bending stress (load 140 mm up)	H × 140 / (πd³/32)	8.2 MPa
FEA peak at the root fillet	hub FEA, 10 node tetrahedra	21.0 MPa, SF 11.9
Crane lift (25 % allowance for slings and dynamic load)	1.25 × 8.89 t	11.1 t

7. Drive (electric actuator)

With the levels equal to within 50 mm, the most water push on a moving leaf is 9.81 × 0.05 × 5.5 × 2.46 = 6.64 kN at mid length, a torque of 8.2 kNm. Pintle friction (μ = 0.1 on a 75 mm radius) adds 0.65 kNm, and water drag at 0.05 m/s is negligible. Design torque 10 kNm. The actuator lever arm about the pivot is 0.85 m shut and 0.90 m open, so the actuator needs **12 kN**. Stroke 1256 mm (pin centres 3301 to 2045 mm): a 20 kN, 1300 mm stroke electric actuator is chosen.

8. Summary

Item	Load or stress	Limit	Safety factor
Skin panel (FEA)	34.1 MPa	355 MPa	10.4
Bottom girder (hand, simply supported)	4.2 MPa	355 MPa	84
Bottom pintle (FEA)	21.0 MPa	250 MPa	11.9
Actuator	12 kN	20 kN	1.7

All checks pass with a safety factor of at least 1.5. Further work for the detailed design: boat impact on the fenders, fatigue of the hinge welds over 100 000 cycles, and a check of the leaf twisting as it swings.